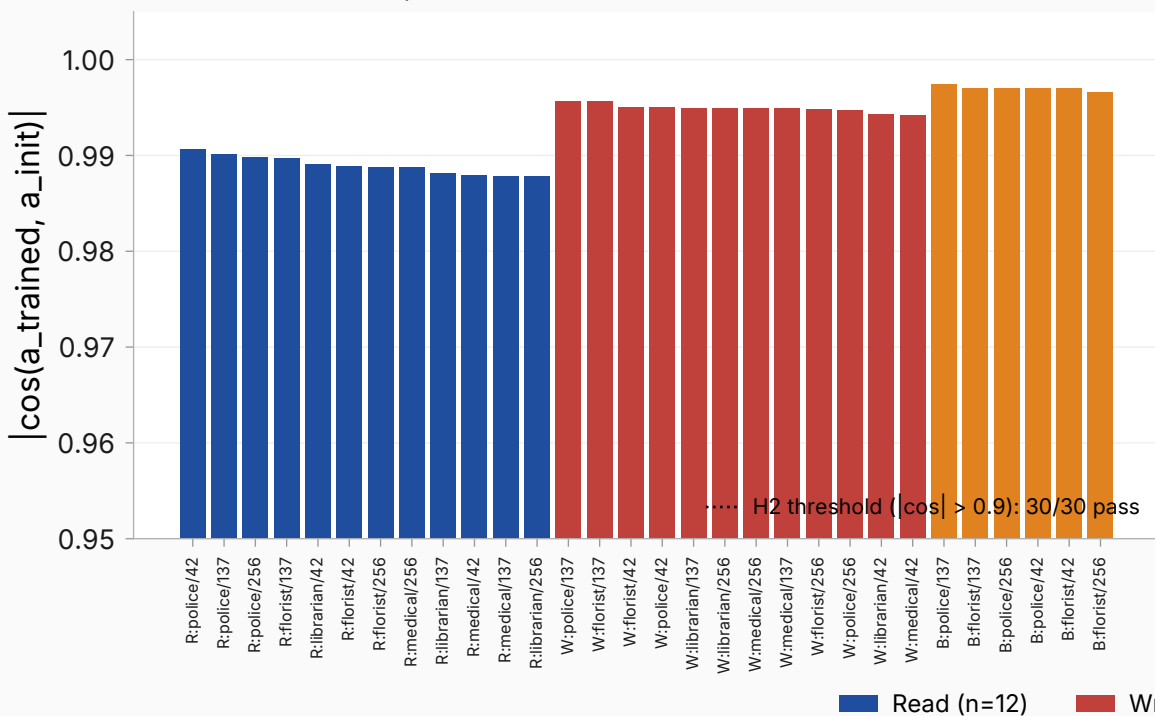


(a) $|\cos(a_{\text{trained}}, a_{\text{init}})| \geq 0.988$ across 30 cells
 median 0.9947 ($\approx 5.9^\circ$), read-arm min 0.9878 ($\approx 8.9^\circ$)



(b) Net A rotation: 23/30 cells below 0.15; 7 read-arm cells at 0.151–0.157
 read median 0.151, write median 0.094, bridge median 0.078

